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DETAIL SPECIFICATION

CHARCOAL (CARBON), ACTIVATED, UNIMPREGNATED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of unimpregnated, activated charcoal (carbon) for use in submarine air filtration, including main fan rooms, sanitary tank and water closet vents, Carbon Dioxide Removal Plants, and Liquid Oxygen/Nitrogen Plants.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Table for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D2854 - Standard Test Method for Apparent Density of Activated Carbon

ASTM D2862 - Standard Test Method for Particle Size Distribution of Granular Activated Carbon

ASTM D2867 - Standard Test Methods for Moisture in Activated Carbon

ASTM D3466 - Standard Test Method for Ignition Temperature of Granular Activated Carbon

ASTM D3802 - Standard Test Method for Ball-Pan Hardness of Activated Carbon

ASTM D5742 - Standard Test Method for Determination of Butane Activity of Activated Carbon

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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ASTM D6556 - Standard Test Method for Carbon Black – Total and External Surface Area by Nitrogen Adsorption

ASTM E11 - Standard Specification for Woven Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Environmentally preferable or biobased materials. Environmentally preferable or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs. Recycled or regenerated activated charcoal (carbon) shall not be used (see 3.3).

3.3 Materials. The activated charcoal (carbon) specified herein shall be a high micropore type suitable for the adsorption of trace organic and inorganic vapors in air. Only new activated charcoal (carbon) shall be furnished. Mercury shall not be used for any purpose. Although not required, activated charcoals (carbons) derived from coconut shell are commonly used for removal of trace contaminants in air.

3.4 Butane activity (adsorption capacity). The activated charcoal (carbon) shall have a butane activity (adsorption capacity) of not less than 19.6 grams of butane per 100 grams of carbon when tested as specified in 4.7.1.

3.5 Surface area. The activated charcoal (carbon) shall have a surface area not less than 1,000 square meters per gram when tested as specified in 4.7.2.

3.6 Particle size distribution. The particle size distribution shall be as specified in table I when tested as specified in 4.7.3.

TABLE I. Particle size distribution.

Sieve(s)	Percent retained
On No. 4	0.0
Through No. 4, on No. 6	0.0 – 5.0
Through No. 6, on No. 12	90.0 – 100.0
Through No. 12, on No. 16	0.0 – 5.0
Through No. 16, on No. 50	0.0 – 0.4
Through No. 50, on pan	0.0 – 0.2

3.7 Ball pan hardness. The ball pan hardness shall be greater than 94.0 percent when tested as specified in 4.7.4.

3.8 Apparent density. The apparent density shall be between 0.42 and 0.57 grams per milliliter when tested as specified in 4.7.5.

3.9 Moisture content. The moisture content shall be less than 5.0 percent when tested as specified in 4.7.6.

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3.10 Ignition temperature. The ignition temperature shall be not less than 350 °C when tested as specified in 4.7.7.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on activated charcoal (carbon) when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table II. When a first article inspection is required, the quantity sampled shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. Testing shall be the same as that which is required for a regular production lot (see 4.4 through 4.8).

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests specified in table II.

TABLE II. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Materials	X	X	3.3	4.9
Butane activity (adsorption capacity)	X	X	3.4	4.7.1
Surface area	X	X	3.5	4.7.2
Particle size	X	X	3.6	4.7.3
Ball pan hardness	X	X	3.7	4.7.4
Apparent density	X	X	3.8	4.7.5
Moisture content	X	X	3.9	4.7.6
Ignition temperature	X	X	3.10	4.7.7

4.4 Lot size.

4.4.1 Activated charcoal (carbon). A lot shall consist of not more than 2,000 pounds of carbon produced by one manufacturing process or in successive batch increments of the same process and the same raw material. If a continuous manufacturing process produces more than 2,000 pounds, a lot shall be defined as each 2,000-pound increment of material produced.

4.4.2 Filled bags. The lot size of filled bags shall consist of each day's production of finished bags or the number presented for the delivery order, whichever is smaller. This does not apply when purchasing activated carbon in drums.

4.5 Sampling.

4.5.1 Sampling for examination of activated charcoal (carbon). From each production lot of activated charcoal (carbon), a representative sample of not less than 5 pounds shall be taken using the sampling thief equipment. The sample shall be thoroughly mixed and placed in a clean, dry container. The container shall have an airtight closure and shall be labeled to identify the lot represented. The contents of each container shall be subjected to the tests specified in 4.7 and the balance of the sample retained until the lot has been determined to be acceptable.

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4.5.2 Sampling for examination of filled bags. A random sample of the filled bags (see table III) shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4, general inspection level II, normal inspection, and the Acceptance Quality Limit (AQL) specified (see 6.2). Table III presents the number of filled bag samples required based on production lot size. This does not apply when purchasing activated carbon in drums.

TABLE III. Sampling for examination of filled bags

Number of bags in lot	Number of bags required to be sampled/tested
2 – 8	2
9 – 15	3
16 – 25	5
26 – 50	8
51 – 90	13
91 – 150	20
151 – 280	32
281 – 500	50
501 – 1,200	80

4.6 Activated charcoal (carbon) tests. Each of the samples selected for testing as specified in 4.5.1 shall be subjected to all of the tests specified in 4.7. Samples shall be divided into portions and all tests shall be conducted twice on each sample. A sample failing any of the tests in 4.7 shall be cause for rejection of the entire activated charcoal (carbon) lot.

4.7 Activated charcoal (carbon) test procedures.

4.7.1 Butane activity (adsorption capacity). The apparatus and test procedures used to determine compliance with the butane activity requirements specified in 3.4 shall be in accordance with ASTM D5742 or equivalent method with written approval from NAVSEA 05 (see 6.7). Use of ASTM D5742 in place of ASTM D3467 (carbon tetrachloride test) is in accordance with most industrial standards and eliminates the use of carbon tetrachloride, which is toxic and an ozone-depleting chemical subject to restriction imposed by the Montreal Protocol. The required result (19.6 grams per 100 grams carbon) is documented in ASTM D5742 to be equivalent to the prior specification of 50 grams of carbon tetrachloride per 100 grams of carbon.

4.7.2 Surface area. The apparatus and test procedures used to determine compliance with the surface area requirements specified in 3.5 shall be in accordance with ASTM D6556 or equivalent method for granular activated charcoal (carbon) with written approval from NAVSEA 05 (see 6.7).

4.7.3 Particle size distribution. The apparatus and test procedures used to determine compliance with the particle size distribution requirements specified in 3.6 shall be in accordance with 4.7.3.1 and 4.7.3.2. These procedures greatly simplify the methods described in ASTM D2862 in the handling and mixing of samples, number of sieves utilized, and required particle size distribution, but utilize similar methodology.

4.7.3.1 Apparatus. The sieves to be employed in conducting the particle size test shall be U.S. Standard testing sieves, or equivalent, conforming to ASTM E11 for sieves for testing purposes. The sieves shall be 2 inches in height and 8 inches in diameter. A standard vibrating machine capable of accommodating six sieves plus a bottom pan and top cover shall be used. An electric timing switch shall also be used.

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4.7.3.2 Procedure. Sieves 4, 6, 12, 16, and 50, plus a bottom pan and top cover, with the largest sieve at the top and the smallest at the bottom, shall be nested. A sample of 100±5 grams of charcoal that meets the moisture content requirements of 3.9 shall be weighed to a precision of at least ±0.1 gram and placed in the top sieve. The nest of sieves, with the cover and pan in place, shall be placed in the testing machine and vibrated for 10 minutes ±10 seconds. The percent retained on each sieve and the pan shall be calculated as follows:

$$\text{Percent retained} = \frac{\text{Weight of charcoal (in pan or sieves)} \times 100}{\text{Sum of individually retained weights}}$$

If the sum of all individually retained weights differs from the original sample of weight by more than 2.0 grams, the test shall be repeated.

4.7.4 Ball pan hardness. The apparatus and test procedures used to determine compliance with the ball pan hardness requirements specified in 3.7 shall be in accordance with ASTM D3802.

4.7.5 Apparent density. The apparatus and test procedures used to determine compliance with the apparent density requirements specified in 3.8 shall be in accordance with ASTM D2854 or equivalent method with written approval from NAVSEA 05 (see 6.7).

4.7.6 Moisture content. The apparatus and test procedures used to determine compliance with the moisture content requirements specified in 3.9 shall be in accordance with ASTM D2867 or equivalent method with written approval from NAVSEA 05 (see 6.7).

4.7.7 Ignition temperature. The apparatus and test procedures used to determine compliance with the ignition temperature requirements specified in 3.10 shall be in accordance with ASTM D3466 or equivalent method with written approval from NAVSEA 05 (see 6.7).

4.8 Examination of filled bags. Each sample of filled bags selected in accordance with 4.5.2 shall be examined for defects of the bag cloth, seams, stitching, and evidence of leakage (see 6.3). Each sample shall be weighed to determine the amount of the contents. Unless otherwise specified (see 6.2), any bag in the sample having one or more defects, or under the required 5.2-pound fill, shall be cause for rejection. This does not apply when purchasing activated carbon in drums.

4.9 Materials. Materials shall be verified as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The primary use of the activated charcoal (carbon) covered by this specification is the removal of a wide range of trace contaminants in the air aboard naval submarines, ensuring the health of the military crew. This function requires high surface area, low pore size activated charcoal (carbon). In addition, the charcoal (carbon) is used in sanitary and water closet vents and for filtration of Carbon Dioxide Removal Plant working fluid. In addition to the 5.2-pound bags, 200-pound drums can be purchased for carbon used with Liquid Oxygen/Nitrogen Plants (see 6.7). Activated charcoal (carbon) procured to this specification is a form, fit, and function replacement for prior revisions. Current stock of activated charcoal (carbon) may be used until depleted; future requisitions for activated charcoal (carbon) should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. AQL of 1.5 percent (see 4.5.2 and 4.8).

d. Requirement for a certificate of materials (see 4.9).

e. Packaging requirements (see 5.1 and 6.3).

f. Level of packing required (see 6.4).

g. Requirements for markings (see 6.5).

6.3 Packaging. The following packaging descriptions (see 6.2) are intended for carbon in 5.2-pound bags and do not apply when purchasing activated carbon in drums.

6.3.1 Weight of activated charcoal (carbon) per bag. A quantity of 5.2+0.2/-0.0 pounds of activated charcoal (carbon) should be furnished in each bag.

6.3.2 Bag cloth. The activated charcoal (carbon) should be supplied in unbleached cotton sheeting cloth bags having dimensions of 12 inches by 8 inches by 5 inches. The physical requirements of the cotton sheeting should be 56±6 by 56±6 yarns-per-inch, a fabric weight of 4.0±0.4 yards-per-pound, with a breaking strength (grab method) of not less than 45 pounds (warp) and 40 pounds (weft/filling).

6.3.3 Bag fabrication. The bags should be fabricated of unbleached cotton sheeting with double-stitched seam type 500 class stitch over the edges of the two plies, followed by an SSa-1-type stitch reinforcement forming a SSa-2-type seam in accordance with ASTM D6193, or equivalent practice, and square corners. The top of the cotton bag should be left open for filling. After each bag has received 5.2 pounds of activated charcoal (carbon), the bag should be stitched closed and the excess part of the bag folded to give minimum dimensions.

6.3.4 Outer bag. The finished filled cloth bag should be further placed in an individual paper or polymer bag of a minimum size with strength to withstand normal abuse. If kraft paper is used, the bag should be double-layer and should be single- or double-row stitched in accordance with ASTM D6193 using stitch type 101 or 301, or equivalent, with seam type BSa-1 or BSa-2, or equivalent, using a binding strip over the material ends. If a polymer bag of sufficient strength is used, the polymer bag should be heat-sealed.

6.4 Packing. Packing should be level A or B as specified (see 6.2) for activated charcoal (carbon) in 5.2-pound bags. The following packaging descriptions do not apply to activated carbon in drums.

6.4.1 Level A. Twelve units of activated charcoal (carbon), packaged as specified in 6.3, should be packed in a fiberboard box conforming to style RSC/0201 or OSC/0202, type CF, variety SW, class WR (weather-resistant), grade V3c of ASTM D5118/D5118M or higher. Standard and WR grade fiberboard stock should conform to ASTM D4727/D4727M. Box sealing/closure, waterproofing, and reinforcing should be in accordance with method V of ASTM D1974/D1974M. Shipping containers should comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

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6.4.2 Level B. Twelve units of activated charcoal (carbon), packaged as specified in 6.3, should be packed in a fiberboard box conforming to style RSC/0201 or OSC/0202, type CF, class domestic, variety SW, grade 250 of ASTM D5118/D5118M or higher. Fiberboard stock should conform to ASTM D4727/D4727M. Box sealing/closure, waterproofing, and reinforcing should be in accordance with method V of ASTM D1974/D1974M. Shipping containers should comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

6.5 Marking. In addition to any special marking required (see 6.2), interior packages and exterior shipping containers should be marked in accordance with MIL-STD-129, DOT Regulations Title 49. Unit packages and exterior shipping containers should include the following markings:

a. "Activated charcoal (carbon), unimpregnated."

b. "Intended use: Naval submarine air purification, sanitary tank and water closet vent filtration, and Carbon Dioxide Removal Plant working fluid filtration." This intended use statement marking is for carbon in 5.2-pound bags only. The intended use marking does not apply to activated carbon in drums.

6.6 Approval of equivalent method. Requests for written approval of use of an equivalent method should be submitted to NAVSEA 05 at commandstandards@navy.mil.

6.7 National stock numbers (NSN). Table IV provides a list of assigned NSNs that correspond to the items covered by this specification. This list may not include all possible NSNs associated with this document.

TABLE IV. Sampling for examination of filled bags.

NSN	Unit of issue
6810-00-726-7944	5.2-lb cloth bag
6810-00-548-3884	200-lb drum

6.8 Subject term (key word) listing.

Activated carbon

Air purification

Butane

Carbon tetrachloride

Surface area

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activities:
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